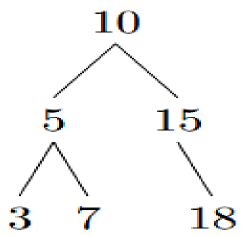
Question 1 — [12 Marks]

You are given the root of a Binary Search Tree (BST) and a value k. Write a function findClosest(root, k) that returns the value in the BST that is closest to k (has the smallest absolute difference with k). If two values are equidistant from k, return the smaller one.

Note: You can use a helper function. You must use recursion. **You cannot add new parameters** to the given function signature and you cannot use global or instance variables. Assume the TreeNode class is already defined with val, left, and right.

Sample Input	Output


public static int findClosest(Node root, int k)
{
   // write your code here
}
-- 	
def findClosest(root, k):
   # write your code here
	findClosest(root, 12)
	Output: 10
	
	Explanation
	Node 10: |12 − 10| = 2 → current closest node is 10
	Node 5: |12 − 5| = 7 → not closer than node 10 difference
	Node 15: |12 − 15| = 3 → not closer than node 10 difference
	Node 3: |12 − 3| = 9 → not closer than node 10 difference
	Node 7: |12 − 7| = 5 → not closer than node 10 difference
	Node 18: |12 − 18| = 6 → not closer than node 10 difference
	
	This is not the exact traversal order of this BST; it is a step-by-step evaluation of all nodes to determine the closest value.

Question 2 — [6 Marks]

2a. In a BST, where is the maximum value always located? [1 Mark]

2b. What is the time complexity of searching in a balanced BST? [1 Mark]

2c. Fill in the blank:
In a max-heap stored as a 1-indexed array, the parent of the node at index ***i*** is at index _______. [1 Mark]

2d. Fill in the blank:
Given the max-heap array [90, 75, 80, 60, 50], after calling extractMax(), the value ______ is placed at the root before heapifying down. [1 Mark]

2e. Which of the following arrays is a valid min-heap? [1 Mark]
(A) [2, 5, 3, 8, 10, 6, 4]   (B) [2, 3, 5, 4, 10, 6, 8]   (C) [2, 5, 3, 4, 10, 6, 8]   (D) [2, 3, 4, 5, 6, 8, 10]

2f. A BST is built by inserting values in this order: [1, 2, 3, 4, 5]. What is the height of the resulting BST? (Height = number of edges on the longest root-to-leaf path.) [1 Mark]